

Measuring error

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Warmup

Your friend collects $n = 100$ observations with a binary response, and fits a logistic regression model:

$$Y_i \sim \text{Bernoulli}(p_i)$$
$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_{99} X_{i,99}$$

Using this data, they then create a confusion matrix:

Predicted Status	Actual Status	
	negative	positive
negative	45	0
positive	0	55

What has gone wrong?

Take-aways from last class activity

- ▶ Even if the model is correctly specified, prediction performance on the **training** data is a biased estimate of prediction performance on **new** data
- ▶ The bias gets worse as the number of explanatory variables in the model increases!

Terminology and notation

Training data:

Fitted model:

New data:

Loss functions

Loss function:

Risk (error)

Training error:

Test error:

Expected test error:

What do we do??

If training error is a **biased** estimate for error on new data, how do we proceed?

Data splitting

- ▶ Divide the available data $\mathcal{T}$ into two disjoint subsets: $\mathcal{T}_{\text{train}}$ and $\mathcal{T}_{\text{test}}$.
- ▶ Common train/test splits include 60/40, 70/30, 80/20

k-fold cross validation

- ▶ Divide the available data $\mathcal{T}$ into k disjoint subsets (called **folds**) $\mathcal{T}_1, \dots, \mathcal{T}_k$

Class activity

Work on the class activity (course website)